

that has to be implemented, and it needs a lot of logic and digital signal processing (DSP) blocks, a high-end FPGA could be used. Even the size of the FPGA is pretty much configurable based on the customer's need.

Traditional analysers have been box type but now analysers are coming with just a single-slot card. Their size is shrinking tremendously, thanks to semi-conductors and multi-core technologies. "FPGA and shrinking size are the key elements bringing down the cost and increasing the capabilities of these instruments," believes Mohanram.

Besides, a single instrument now includes, for instance, spectrum analyser, logic analyser, oscilloscope, digital multimeter and multiple-slot PXI chassis capabilities, which reduces investment on instruments. It also minimises the effort on measurement, automation and learning how to use the instruments, besides occupying much lesser space in labs. Adesh Kumar Jain, applications engineer, Keysight Technologies India Pvt Ltd, says, "Software tools control different instruments from a single software on a single screen. Even mobile versions of software tools are launched by companies to control/monitor measurements from smartphones. These are supplemented with tools to make automation easier."

Also, many power applications have fast-changing asynchronous current pulses which are not suited to fixed data length FFT analysis. Hence, gapped analysis causes events to be missed, resulting in inaccurate measurements. "The Newtons4th Ltd (N4L) power analysers use real-time discrete Fourier transform (DFT) technique with variable-window no-gap analysis to ensure the optimum speed and accuracy at all times," says Motiwale.

Why FPGA based analysers

Apart from the flexibility, FPGA based analysers are better for hardware in-loop testing and test coverage in time. Today, systems are all interconnected. There is lot of security and logic built into every communication protocol to ensure that no signal or information

Other factors shaping analysers

Broadband. LTE, Bluetooth, Wi-Fi, digital TV all require broadband real-time spectrum analysers when signal demodulation is needed.

Extensibility. It should be possible to accommodate continuously-changing broadcast requirements with the same hardware platform using task-specific software written to address the new needs. What could be more cost-effective than having an application programming interface (API) to write new functions?

Easily upgradable. It should be possible to upgrade the analyser easily when the need arises, just as a PC can be upgraded when more memory or storage is required.

—Bruce Devine, CEO, Test Equipment Plus, Inc.

is lost when it goes from one end to another.

Consider people who are building multiple-input and multiple-output (MIMO) systems to define the next generation of mobile communication called 5G. One of the biggest challenges that engineers face is having to define and implement the base station or customer premises equipment (CPE) even before the standard is defined. "In this case, you need logic analysers, spectrum or RF signal analysers, which can really behave as though it's a physical phone, and at the same time analyse the complete protocol that goes between the base station and the CPE. So the protocol is being defined and at the same time you are trying to emulate a particular device. This is not possible with traditional spectrum, logic and RF analysers," informs Mohanram. With FPGA technology, it becomes possible.

Testing is becoming a lot more thorough. According to Mohanram, "Earlier perhaps people used to test four different points, which is a small example, on a protocol to say it is fine. But today, instead of testing limited number of points, they could test pretty much all different points on the curve. So very thorough testing is possible because of FPGA." FPGA is basically a concurrent logic; everything happens in parallel and, because of that, the set time also drastically comes down.

Selecting the right analyser

Flexibility is one factor that is often missed out while selecting analysers. Scientists and engineers are comfortable using traditional instruments and have stuck on to those. It is a mindset people have stuck on to, but now it is

What is CPE?

In 3G and 4G, a base station is the cellular tower and CPE, in this case, is your phone. But moving towards 4G or 5G, instead of a phone it could be a router or a much more sophisticated device which could handle voice, audio, video and may be even some control signals. Mobile phone is one type of CPE that could be a much more complex CPE in upcoming next-generation technologies. Therefore, instead of calling it a mobile phone, it has a more generic name: CPE.

slowly changing. People realise, if they have the capability to define exactly what the analyser can do, they really are able to make optimal use and get the best throughput out of the system or the instrument they are building.

Mohanram says, "What we generally ask them is, 'Tell us what you exactly want to do with an analyser?' and we make them do it right in front of us by guiding them on how they could go about bending the instrument according to their requirements rather than adjusting with what the vendor typically provides." He adds, "Because of the buzz surrounding defining functionality of these instruments, that mindset is slowly changing, and I think that's one point I would like to emphasize which people should keep in mind while selecting these products."

Another factor that people generally ask while selecting analysers is, "Can this analyser analyse this particular protocol?" If it is followed by a yes, they immediately go for it, informs Mohanram. But the real question should be, "Can it be done in real time?" This is not considered while picking analysers. Real-time analysis helps users to do a two-way communication analysis, be it logic or communication from a RF stand point. People forget to ask about